

# Problem Statement #37 — Group Travel Expense Simplification Engine

## 1. Functional Requirements (FR)

| Req ID | Type       | Description                                                                                                                                                                                                                             | Priority | Acceptance Criteria                                                                                                                                                                                        | Rationale                                                                                          | Comments                      |
|--------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------|
| FR-001 | Functional | The system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions.                                                                                               | High     | Pass: Group balance settles to net zero with minimal payment edges; Fail: Sum of debts does not equal sum of credits.                                                                                      | Core value proposition — reduces the number of payments travelers must make to settle up.          | Given                         |
| FR-002 | Functional | The system shall allow a Group Traveler to add a shared expense and specify an uneven split (custom amount or percentage) across selected trip members.                                                                                 | High     | When an expense is submitted with a custom split, each traveler's owed amount is recalculated and displayed, and the sum of all shares equals the total expense amount.                                    | Uneven splitting is the core input needed before any settlement can be computed.                   | Given                         |
| FR-003 | Functional | The system shall convert an expense entered in a foreign currency to the trip's base currency using the exchange rate published for that day.                                                                                           | High     | When an expense is logged in a non-base currency, the converted base-currency amount is stored and displayed alongside the original amount and the rate used.                                              | Multi-currency trips cannot be settled correctly without normalizing all expenses to one currency. | Given                         |
| FR-004 | Functional | The system shall allow a Trip Admin to add or remove Group Travelers from a trip.                                                                                                                                                       | Medium   | When a Trip Admin removes a traveler with a zero balance, the traveler no longer appears in the trip's member list or future expense splits.                                                               | Trip membership must be managed before expenses can be attributed correctly.                       | Given                         |
| FR-005 | Functional | The system shall allow a Group Traveler to view their real-time net balance (amount owed or owed to them) for a trip.                                                                                                                   | Medium   | When a traveler opens the balance screen, the displayed net amount matches the sum of their expense shares minus payments made, recalculated within 2 seconds of the last update.                          | Travelers need visibility into their balance before and after settlement.                          | Given                         |
| FR-006 | Functional | The system shall allow a Group Traveler to confirm one or more settlement payments as completed, update each affected traveler's balance, and mark the corresponding settlement edge as paid.                                           | High     | Pass: After confirmation, payer and payee balances change by exactly the payment amount and the edge shows "Paid"; Fail: balances or edge status unchanged or inconsistent.                                | Closes the settlement loop and keeps balances accurate as payments happen outside the app.         | Derived: UC steps 6–7         |
| FR-007 | Functional | The system shall display, on "Settle Up", all recorded trip expenses and each traveler's current net balance, and shall present the proposed settlement plan stating who pays whom and how much.                                        | High     | Pass: Every expense and balance is listed; each plan edge shows payer, payee and amount; each debtor's outgoing payments sum to their balance. Fail: any expense or edge missing.                          | Travelers must be able to review and trust the plan before paying.                                 | Derived: UC steps 1, 4, 5     |
| FR-008 | Functional | The system shall aggregate all expense shares and payments to compute each traveler's net balance in the trip's base currency.                                                                                                          | High     | Pass: Net balance = total paid – total owed share for every traveler, and the sum of all net balances is zero ( $\pm 0.01$ ). Fail: any mismatch beyond tolerance.                                         | Net balances are the input to the settlement algorithm (FR-001).                                   | Derived: UC step 2            |
| FR-009 | Functional | If the exchange-rate service is unreachable, the system shall fall back to the last successfully cached rate, flag the affected expenses as "estimated", and display a warning banner while proceeding with the settlement calculation. | High     | Pass: With the rate service disabled, a settlement is still generated, affected expenses carry an "Estimated" flag and a banner is shown. Fail: settlement is blocked or estimated expenses are unflagged. | An external outage must not block settlement, but users must know the figures are provisional.     | Derived: UC alt. flow 3a1–3a3 |
| FR-010 | Functional | When live exchange rates become available again, the system shall recalculate the settlement for estimated expenses and notify affected travelers of any change.                                                                        | Medium   | Pass: After service recovery, estimated flags are cleared, the settlement is recomputed and each traveler whose amount changed is notified. Fail: stale estimate retained or no notification sent.         | Ensures final amounts reflect real rates and travelers are aware of changes.                       | Derived: UC alt. flow 3a4     |

| Req ID | Type       | Description                                                                                                                                                                                                                                                                                 | Priority | Acceptance Criteria                                                                                                                                                                       | Rationale                                                                          | Comments                       |
|--------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------|
| FR-011 | Functional | The system shall notify Group Travelers when a settlement plan is generated, when a payment is confirmed, and when a plan changes after recalculation.                                                                                                                                      | Medium   | Pass: Every affected traveler receives an in-app/email notification for each event within 60 seconds. Fail: any affected traveler not notified.                                           | Keeps all travelers informed without manual follow-up (Notify Travelers use case). | Derived: Use Case Diagram      |
| FR-012 | Functional | The system shall confirm that a trip is fully settled once all settlement edges are marked paid.                                                                                                                                                                                            | Medium   | Pass: When the last edge is marked paid, all balances are zero and the trip status is "Settled". Fail: status "Settled" while any edge is unpaid or any balance is non-zero.              | Gives travelers a clear end state for the trip.                                    | Derived: UC step 8             |
| FR-013 | Functional | The system shall verify that the trip has at least two Group Travelers, at least one expense and all expenses normalized to the base currency before settlement; if verification or calculation fails, no settlement graph shall be generated and existing balances shall remain unchanged. | Medium   | Pass: Settle Up on a trip with one traveler or no expenses returns an explanatory error, no graph is stored and balances are unchanged. Fail: a graph is created or balances are altered. | Enforces the use-case preconditions and failure postcondition.                     | Derived: UC pre/postconditions |

## 2. Non-Functional Requirements (NFR)

| Req ID  | Type           | Description                                                                                                                                                                                                                                            | Priority | Acceptance Criteria                                                                                                                                                                                            | Rationale                                                                                                               | Comments            |
|---------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------|
| NFR-001 | Non-functional | The settlement calculation engine must execute in under 100 ms for groups of up to 50 travelers with 500 expenses.                                                                                                                                     | High     | Pass: Benchmarking tests confirm target latency under simulated peak load.                                                                                                                                     | Ensures the app remains responsive even for large group trips.                                                          | Given               |
| NFR-002 | Non-functional | The system shall encrypt all stored and transmitted financial data (expenses, balances, payment details) using industry-standard encryption (e.g., AES-256 at rest, TLS 1.2+ in transit).                                                              | High     | Pass: Security audit / automated scan confirms all financial data fields are encrypted at rest and all API traffic uses TLS 1.2 or higher; Fail: any financial field found in plaintext in storage or transit. | Financial and personal data must be protected against breaches given the sensitive nature of group payment information. | Provided            |
| NFR-003 | Non-functional | The balance view and the Settle Up screen shall load and refresh within 2 seconds (95th percentile) after any expense or payment change, under a load of 500 concurrent users.                                                                         | Medium   | Pass: Load test shows p95 response $\leq 2$ s at 500 concurrent users; Fail: p95 exceeds 2 s.                                                                                                                  | Supports the real-time balance requirement (FR-005).                                                                    | Derived from FR-005 |
| NFR-004 | Non-functional | The system shall be available at least 99.5% per month (excluding planned maintenance) with a recovery point objective $\leq 5$ minutes and a recovery time objective $\leq 1$ hour.                                                                   | High     | Pass: Monitoring shows $\geq 99.5\%$ uptime over 30 days; a restore drill meets RPO/RTO. Fail: either target missed.                                                                                           | Travelers settle up at the end of trips, often at fixed times, so downtime or data loss is costly.                      | —                   |
| NFR-005 | Non-functional | All monetary amounts shall be stored as fixed-point decimals (minor units) with a single documented rounding rule; rounding residuals shall be allocated so that shares always sum to the expense total and all balances net to zero after settlement. | High     | Pass: Property-based tests over 10,000 random scenarios show zero discrepancy greater than 0.01; Fail: any scenario leaves a non-zero residual.                                                                | Floating-point errors would break the "net balance equals zero" guarantee.                                              | —                   |
| NFR-006 | Non-functional | The system shall enforce role-based access control: only a Trip Admin may add or remove travelers, and a user may view or modify only trips of which they are a member.                                                                                | High     | Pass: Authorization tests show unauthorized requests return HTTP 403 and leak no trip data; Fail: any cross-trip or cross-role access succeeds.                                                                | Protects privacy and prevents unauthorized changes to membership and balances.                                          | —                   |
| NFR-007 | Non-functional | Calls to the external exchange-rate service shall use a timeout of 3 s, bounded retries with back-off and a circuit breaker, and the system shall keep all other functions available during an outage.                                                 | High     | Pass: Fault-injection test with the service down shows add-expense, balance and settlement still work using cached rates; Fail: any core function errors.                                                      | Supports the fallback behaviour in FR-009 and avoids cascading failures.                                                | —                   |
| NFR-008 | Non-functional | The system shall scale horizontally to support at least 1,000 concurrent trips and 10,000 concurrent users without code changes.                                                                                                                       | Medium   | Pass: Load test at target volume shows throughput grows with added instances and error rate $< 1\%$ .                                                                                                          | Allows growth beyond the initial user base without redesign.                                                            | —                   |
| NFR-009 | Non-functional | The system shall write an immutable audit record (actor, timestamp, before/after values) for every expense, membership and payment-status change and retain it for at least 12 months.                                                                 | Medium   | Pass: Sampled changes each produce a complete audit entry that cannot be edited or deleted by application users; Fail: missing or mutable entries.                                                             | Resolves disputes about who changed what in a shared-money context.                                                     | —                   |
| NFR-010 | Non-functional | The user interface shall be responsive from 360 px screen width upwards, support the latest two versions of Chrome, Safari, Firefox and Edge, and meet WCAG 2.1 AA.                                                                                    | Medium   | Pass: Cross-browser/device test passes on the listed targets and an accessibility scan reports no AA violations; $\geq 80\%$ of first-time test users add an expense unaided in under 60 s.                    | Most travelers use phones while travelling and must be able to use the app without training.                            | —                   |
| NFR-011 | Non-functional | The settlement engine and currency-conversion modules shall be isolated behind defined interfaces and maintain at least 80% automated unit-test coverage, enforced by the CI pipeline.                                                                 | Medium   | Pass: CI fails the build when coverage drops below 80% or interface contract tests fail.                                                                                                                       | Allows the algorithm and rate provider to be replaced or upgraded safely.                                               | —                   |

### 3. Requirements Traceability Matrix (RTM)

Traces each requirement to its use case and flow step, the architecture component(s) that realise it, and the verification that will prove it. Status values: Planned → In Design → Implemented → Verified.

| Req ID  | Requirement (short)                              | Use Case                                               | Flow Reference                       | Actor(s)                         | Architecture Component(s)                                                            | Test Case ID | Verification           | Status  |
|---------|--------------------------------------------------|--------------------------------------------------------|--------------------------------------|----------------------------------|--------------------------------------------------------------------------------------|--------------|------------------------|---------|
| FR-001  | Compute min-flow settlement graph                | Compute Debt Settlement                                | Main flow step 3                     | Group Traveler                   | Settlement Engine; PostgreSQL                                                        | TC-FR-001    | Test                   | Planned |
| FR-002  | Add shared expense with uneven split             | Add Expense                                            | Use Case Diagram (flow not detailed) | Group Traveler                   | Web Client; API Gateway; Expense Service; PostgreSQL                                 | TC-FR-002    | Test                   | Planned |
| FR-003  | Convert foreign currency to base currency        | Convert Currency (included by Add Expense)             | Precondition 3; Diagram «include»    | Group Traveler, Currency Service | Currency Conversion Service; Exchange-Rate Adapter; Redis Cache                      | TC-FR-003    | Test                   | Planned |
| FR-004  | Add / remove trip travelers                      | Manage Trip Travelers                                  | Use Case Diagram (flow not detailed) | Trip Admin                       | Web Client; Trip & Member Service; PostgreSQL                                        | TC-FR-004    | Test                   | Planned |
| FR-005  | View real-time net balance                       | View Balance                                           | Main flow step 1–2                   | Group Traveler                   | Web Client; Balance Service; Redis Cache                                             | TC-FR-005    | Test                   | Planned |
| FR-006  | Confirm payments, mark edges paid                | Compute Debt Settlement                                | Main flow step 6–7                   | Group Traveler                   | Payment Tracking Service; Balance Service; PostgreSQL                                | TC-FR-006    | Test                   | Planned |
| FR-007  | Show expenses, balances and settlement plan      | Compute Debt Settlement                                | Main flow step 1, 4, 5               | Group Traveler                   | Web Client; Expense Service; Settlement Engine                                       | TC-FR-007    | Demonstration          | Planned |
| FR-008  | Aggregate net balances                           | Compute Debt Settlement                                | Main flow step 2                     | Group Traveler                   | Balance Service; PostgreSQL                                                          | TC-FR-008    | Test                   | Planned |
| FR-009  | Cached-rate fallback and "estimated" flag        | Compute Debt Settlement                                | Alt. flow 3a1–3a3                    | Group Traveler, Currency Service | Currency Conversion Service; Exchange-Rate Adapter; Redis Cache; Web Client (banner) | TC-FR-009    | Test (fault injection) | Planned |
| FR-010  | Recalculate and notify on live-rate recovery     | Compute Debt Settlement; Notify Travelers              | Alt. flow 3a4                        | Group Traveler, Currency Service | Currency Conversion Service; Settlement Engine; Event Bus; Notification Service      | TC-FR-010    | Test                   | Planned |
| FR-011  | Notify travelers of plan and payment events      | Notify Travelers (included by Compute Debt Settlement) | Diagram «include»                    | Group Traveler                   | Notification Service; Event Bus; Notification Adapter                                | TC-FR-011    | Test                   | Planned |
| FR-012  | Confirm trip fully settled                       | Compute Debt Settlement                                | Main flow step 8                     | Group Traveler                   | Payment Tracking Service; Balance Service                                            | TC-FR-012    | Test                   | Planned |
| FR-013  | Validate preconditions; no graph on failure      | Compute Debt Settlement                                | Preconditions; Failure postcondition | Group Traveler                   | Settlement Engine; Expense Service; Trip & Member Service                            | TC-FR-013    | Test                   | Planned |
| NFR-001 | Settlement < 100 ms (50 travelers, 500 expenses) | Compute Debt Settlement                                | Main flow step 3                     | Group Traveler                   | Settlement Engine                                                                    | TC-NFR-001   | Performance benchmark  | Planned |
| NFR-002 | AES-256 at rest, TLS 1.2+ in transit             | All use cases                                          | All flows                            | All                              | API Gateway; PostgreSQL; Redis Cache; Adapters; KMS                                  | TC-NFR-002   | Security scan / audit  | Planned |

| Req ID  | Requirement (short)                              | Use Case                                                    | Flow Reference                         | Actor(s)                   | Architecture Component(s)                                        | Test Case ID | Verification                   | Status  |
|---------|--------------------------------------------------|-------------------------------------------------------------|----------------------------------------|----------------------------|------------------------------------------------------------------|--------------|--------------------------------|---------|
| NFR-003 | Balance and Settle Up views ≤ 2 s (p95)          | View Balance; Compute Debt Settlement                       | Main flow step 1–2                     | Group Traveler             | Balance Service; Redis Cache; API Gateway                        | TC-NFR-003   | Load test                      | Planned |
| NFR-004 | 99.5% availability; RPO ≤ 5 min, RTO ≤ 1 h       | All use cases                                               | All flows                              | All                        | PostgreSQL (replica, backups); stateless services; health checks | TC-NFR-004   | Monitoring / DR drill          | Planned |
| NFR-005 | Fixed-point money, zero residual balances        | Add Expense; Compute Debt Settlement                        | Main flow step 2–3                     | Group Traveler             | Expense Service; Balance Service; Settlement Engine              | TC-NFR-005   | Property-based test            | Planned |
| NFR-006 | Role-based access control                        | Manage Trip Travelers; all trip data access                 | All flows                              | Group Traveler, Trip Admin | API Gateway (authN/authZ); Trip & Member Service                 | TC-NFR-006   | Security test                  | Planned |
| NFR-007 | Timeout, retry, circuit breaker for rate service | Convert Currency                                            | Alt. flow 3a                           | Currency Service           | Exchange-Rate Adapter; Redis Cache                               | TC-NFR-007   | Fault-injection test           | Planned |
| NFR-008 | Horizontal scale: 1,000 trips / 10,000 users     | All use cases                                               | All flows                              | All                        | Stateless services; API Gateway; Event Bus; PostgreSQL           | TC-NFR-008   | Load test                      | Planned |
| NFR-009 | Immutable audit trail, 12-month retention        | Add Expense; Manage Trip Travelers; Compute Debt Settlement | Main flow step 6–7; membership changes | Group Traveler, Trip Admin | Audit Log Store; Event Bus                                       | TC-NFR-009   | Inspection / test              | Planned |
| NFR-010 | Responsive, cross-browser, WCAG 2.1 AA           | All use cases                                               | All flows                              | All                        | Web / Mobile Client                                              | TC-NFR-010   | Usability + accessibility test | Planned |
| NFR-011 | Modular design, ≥ 80% unit-test coverage         | Compute Debt Settlement; Convert Currency                   | Main flow step 3; 3a                   | Dev team                   | Settlement Engine; Currency Conversion Service; CI pipeline      | TC-NFR-011   | Inspection / CI report         | Planned |

## Use-case coverage view

| Use Case                                               | Actor(s)                     | Requirements covering it                                                         |
|--------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------|
| Add Expense                                            | Group Traveler               | FR-002, FR-003 (include), FR-007, NFR-005, NFR-009                               |
| Convert Currency                                       | Currency Service (secondary) | FR-003, FR-009, FR-010, NFR-007                                                  |
| Compute Debt Settlement                                | Group Traveler, Trip Admin   | FR-001, FR-006, FR-007, FR-008, FR-009, FR-010, FR-012, FR-013, NFR-001, NFR-005 |
| Notify Travelers (included by Compute Debt Settlement) | Group Traveler               | FR-010, FR-011                                                                   |
| Manage Trip Travelers                                  | Trip Admin                   | FR-004, NFR-006, NFR-009                                                         |
| View Balance                                           | Group Traveler               | FR-005, FR-008, NFR-003                                                          |

## 5. Key Architecture Decisions

- Layered, service-oriented design: client → API gateway → domain services → data/messaging → integration adapters. Each domain service maps to one or more use cases, so the RTM traces cleanly to components.
- Settlement Engine is a pure, stateless module that takes net balances and returns a minimal set of payment edges (min-flow). Isolating it meets the 100 ms target (NFR-001) and makes it easy to test and swap (NFR-011).
- Currency conversion is separated from expense entry (the «include» relationship). The Exchange-Rate Adapter wraps the external Currency Service with timeout, retry and a circuit breaker, and the Redis cache holds the last good rate so settlement continues in estimated mode (FR-009, NFR-007).
- Event bus decouples notifications and recalculation: RateUpdated triggers re-settlement (FR-010), and SettlementComputed / PaymentConfirmed drive notifications (FR-011) without blocking the user request.
- PostgreSQL is the system of record (ACID transactions keep payments and balances consistent). Redis caches balances and rates only, so losing it affects speed, not correctness.
- Security: TLS 1.2+ on every hop, AES-256 at rest with keys in a KMS, JWT authentication and role-based authorization at the gateway, and an append-only audit log (NFR-002, NFR-006, NFR-009).
- Statelessness of the services layer allows horizontal scaling behind the gateway (NFR-008) and supports the availability target (NFR-004).